SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – JIGSAW ACTIVITY

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO4 – Articulation (BL3)	**Assessment:**	Assessment Tool 2 – Jigsaw Activity
Weightage:	35% of CIA	**Total Marks:**	100
CO4 Statement:	*Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data.*		
Student Name:	*Pugazharasan K*	**Reg. No.:**	*192511341*
Section / Batch:	*B*	**Date:**	19/08/2024

Code of Conduct

I ***Pugazharasan K (192511341)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: *Pugazh.K*

Instructions

- Home Groups (8): 5 students, Expert Groups: One expert per topic. Total: 100 Marks.
- Students first discuss in Expert Groups. They return to Home Groups and teach teammates.
- Each student teaches Concept, Working, Advantages, Disadvantages, Applications, Time Complexity.

Section / Topic	Focus	Q Nos	Marks
Hashing & Sorting	Hashing, Resolution Techniques, Sorting	8 Teams	100
		GRAND TOTAL:	**100 Marks**

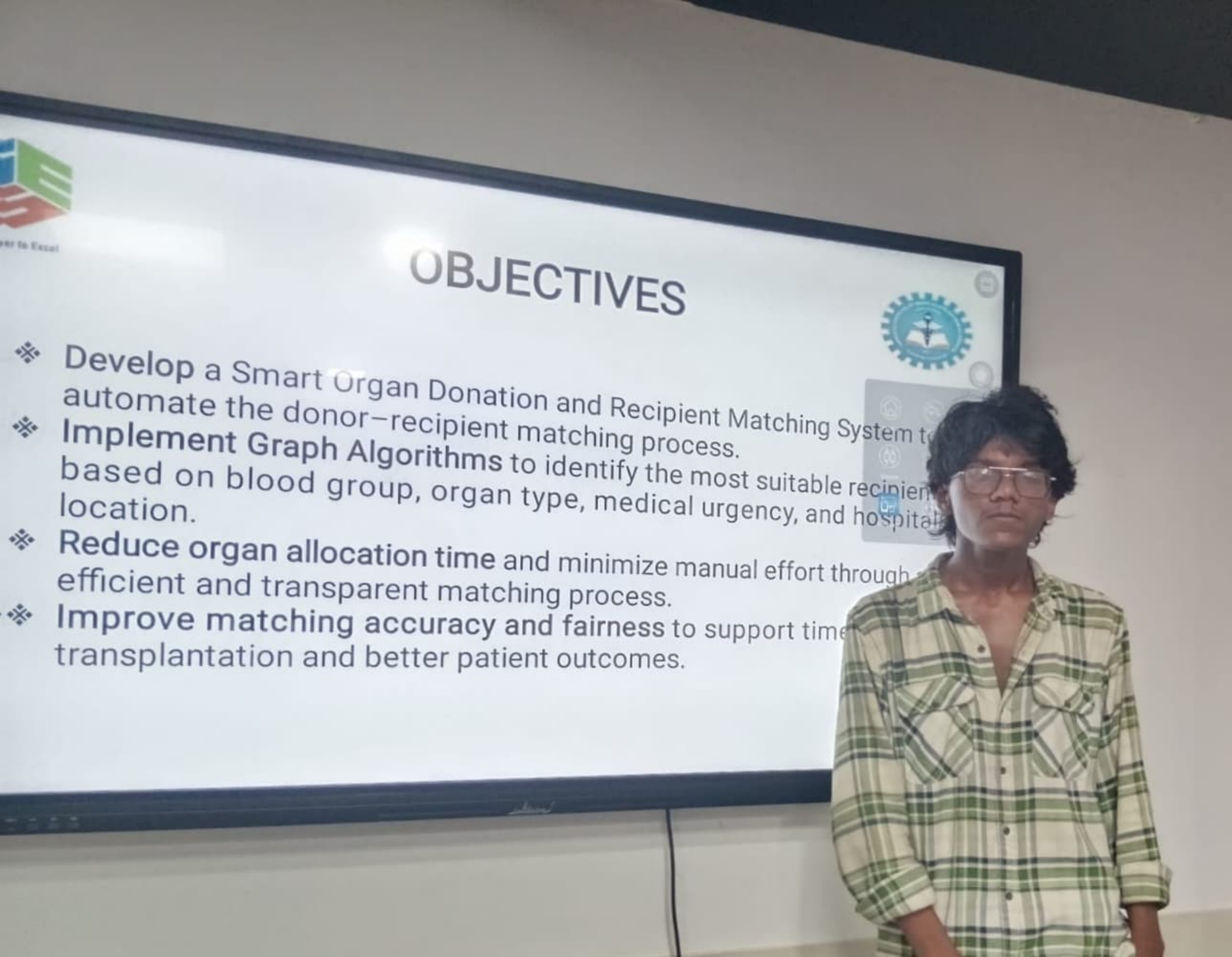
Annexure A – Jigsaw Activity

1. Scenario 1 – Airport Baggage System (Every luggage has Barcode)
    a. Hashing Method
    b. Collision handling
    c. Fast luggage
2. Scenario 2 – E-Commerce Warehouse (Products stored by SKU)
    a. Hashing strategy
    b. Inventory retrieval
    c. Sorting inventory.
3. Scenario 3 – Traffic Police Database (Vehicle Registration Numbers)
    a. Fast searching
    b. Duplicate handling
    c. Sorting violations
4. Scenario 4 – Social Media Platform (Millions of users)
    a. Hash Function
    b. Collision handling – Separate Chaining
    c. Fast searching
5. Scenario 19 – University Admission Portal (A university receives 75,000 applications. Applicants are searched using Application IDs, while merit lists are prepared based on entrance scores.)
6. Scenario 22 – Pharmacy Inventory Management (A pharmacy maintains thousands of medicine records. Pharmacists search medicines by code, while stock reports are arranged based on expiry dates.)
7. Scenario 24 – Weather Monitoring System (A weather station collects data from thousands of sensors every minute. Scientists retrieve records using sensor IDs and generate reports sorted by temperature and rainfall.)
8. Scenario 6 – Digital Music Streaming App (Songs, Artist, Playlist, Genre)
    a. Fast searching
    b. Hash Function
    c. Fast Sorting

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Accurately interprets all problem requirements and constraints before coding.	Understands most requirements with minor omissions.	Partially understands the problem; misses some constraints.	Misinterprets the problem or ignores key requirements.
Correctness of Solution	30	Program compiles successfully and passes all hidden and visible test cases.	Program passes most test cases with minor errors.	Program passes some test cases but has logical issues.	Program fails to compile or produces incorrect results.
Data Structure & Algorithm Selection	20	Selects the most appropriate data structure and algorithm with an optimized solution.	Chooses an appropriate solution with minor inefficiencies.	Uses a workable but less efficient approach.	Uses an inappropriate data structure or algorithm.
Code Quality & Documentation	20	Well-structured, modular, readable, and properly indented code with meaningful identifiers.	Readable code with minor formatting issues.	Basic coding style with inconsistent formatting.	Poorly organized code that is difficult to understand.
Time & Space Optimization	10	Demonstrates excellent optimization and correctly analyses time and space complexity.	Good optimization with minor inefficiencies.	Limited optimization and incomplete complexity analysis.	No optimization and incorrect or missing complexity analysis.

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: _______________	Date: _______________	Date: _______________


OBJECTIVES
Develop a Smart Organ Donation and Recipient Matching System
automate the donor–recipient matching process.
Implement Graph Algorithms to identify the most suitable
based on blood group, organ type, medical urgency, and
location.
Reduce organ allocation time and minimize manual effort through
efficient and transparent matching process.
Improve matching accuracy and fairness to support
transplantation and better patient outcomes.